

Independent Reading Report: Learning-Based Perception, Planning, and Control for Autonomous Robotic Systems

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1 Reading Motivation

This reading project aims to understand the research foundations of autonomous robotic systems, with a particular focus on learning-based navigation, multi-robot coordination, active perception and semantic mapping, open-world task planning, and imitation-based robot manipulation. The five selected papers form a coherent technical path: first, converting raw geometric observations into safe robot motion; second, extending navigation from a single robot to decentralized multi-robot systems; third, actively selecting sensing actions to improve semantic scene understanding; fourth, connecting high-level symbolic planning with language-based reasoning; and finally, learning coordinated manipulation policies from demonstrations.

My own background is in embedded robotics, state estimation, sensor fusion, real-time control, multi-agent decision-making, and robot learning. In the Fast Robots project at Cornell University, I worked with a small autonomous mobile robot using IMU and ToF sensing, Kalman-filter-based distance estimation, yaw estimation, PID control, mapping, localization, BLE-based logging, and autonomous maneuver execution. In the Overcooked-AI project, I explored cooperative multi-agent decision-making, task allocation, navigation, object handling, and shared-space coordination. I also have experience in visual perception through an improved YOLOv11 object detection project and in multi-agent path planning through a publication on high-density automated storage and retrieval systems.

These experiences motivated me to read papers related to Prof. Jia Pan's research on robotics and artificial intelligence. Instead of treating the papers as isolated technical works, I use this reading project to connect my prior hands-on robotics experience with broader research questions in autonomous robot perception, planning, coordination, and real-world deployment.

2 Selected Papers

No.	Paper	Main Topic	Connection to My Background
1	<i>NeuPAN: Direct Point Robot Navigation With End-to-End Model-Based Learning</i>	Direct point-based robot navigation	Fast Robots, ToF/IMU sensing, control
2	<i>Distributed Multi-Robot Collision Avoidance via Deep Reinforcement Learning for Navigation in Complex Scenarios</i>	Decentralized multi-robot navigation	Overcooked-AI, AS/RS path planning
3	<i>Semantics-Aware Receding Horizon Planner for Object-Centric Active Mapping</i>	Semantic active mapping	Fast Robots mapping, YOLO perception
4	<i>Language-Augmented Symbolic Planner for Open-World Task Planning</i>	Open-world task planning	Overcooked-AI, embedded AI command system
5	<i>BiKC+: Bimanual Hierarchical Imitation With Keypose-Conditioned Coordination-Aware Consistency Policies</i>	Bimanual imitation learning	Robot Learning, embodied coordination

3 Paper 1: NeuPAN: Direct Point Robot Navigation With End-to-End Model-Based Learning

Core Idea

The first paper studies direct robot navigation in cluttered and unknown environments. Traditional navigation pipelines often decompose the problem into perception, mapping, planning, and control. Although this modular structure is intuitive, errors in intermediate representations can propagate to downstream planning and control. For example, raw point observations may first be converted into an occupancy map or simplified obstacle model, and approximation errors in this step may affect the final motion command.

NeuPAN addresses this issue by directly using raw point cloud observations for robot navigation. The key idea is to map raw obstacle points into a latent distance feature space and then generate collision-free robot motion through an end-to-end model-based learning framework. Instead of relying only on a black-box neural network, the method is built around a neural proximal alternating-minimization network. This gives the system both learning capacity and model-based interpretability.

The paper is important because it shows a middle path between classical optimization-based navigation and purely data-driven robot policies. Classical methods are interpretable but may be inefficient or overly dependent on hand-designed representations. Purely learned policies can be flexible but may lack physical interpretability. NeuPAN tries to preserve the structure of model-based navigation while using neural learning to improve efficiency and adaptability.

Takeaway

My main takeaway is that robot navigation can be made more robust by reducing unnecessary intermediate representations while still preserving physical and optimization-based structure. This directly connects to my Fast Robots project. In that project, my robot used a modular pipeline: ToF and IMU sensing, Kalman filtering, yaw estimation, PID control, and behavior execution. This pipeline worked, but it also made me aware that each module can introduce errors that affect later modules.

NeuPAN helped me understand a more research-oriented formulation of the same problem: how raw geometric observations can be converted into safe and feasible robot motion. It also suggests a possible future direction for small embedded robots: adapting direct point-based navigation to sparse ToF, IMU, or low-cost depth observations rather than assuming dense LiDAR sensing.

4 Paper 2: Distributed Multi-Robot Collision Avoidance via Deep Reinforcement Learning for Navigation in Complex Scenarios

Core Idea

The second paper studies decentralized multi-robot collision avoidance. In centralized multi-robot navigation, a central server plans paths for all robots using global information. This can provide strong performance in structured settings, but it is difficult to scale to large robot teams and depends heavily on reliable communication and complete knowledge of the environment.

The paper proposes a fully decentralized, sensor-level collision-avoidance policy. Each robot makes its own navigation decision using local observations, including 2D LiDAR measurements, relative goal position, and current velocity. The policy directly maps these observations to velocity commands:

$$a^t = [v^t, \omega^t],$$

where v^t is the translational velocity and ω^t is the rotational velocity. The same policy is shared by all robots, but each robot executes it independently according to its own local observation.

A key contribution is the multi-scenario multi-stage reinforcement learning framework. The policy

is trained in simulation using Proximal Policy Optimization across many robots and diverse environments. The paper also introduces a hybrid control architecture. In simple scenarios, a PID controller moves the robot efficiently toward its goal; in complex scenarios, the learned RL policy handles collision avoidance; in emergent scenarios, a conservative safe policy is used:

$$\pi_{\text{hybrid}} = \begin{cases} \pi_{\text{PID}}, & \text{simple scenario,} \\ \pi_{\text{RL}}, & \text{complex scenario,} \\ \pi_{\text{safe}}, & \text{emergent scenario.} \end{cases}$$

This hybrid design improves real-world robustness and reduces the risk of relying only on a black-box learned policy.

Takeaway

This paper helped me connect multi-agent decision-making with real robotic navigation. In my Overcooked-AI project, I studied two agents coordinating task allocation, navigation, object handling, and delivery under shared spatial constraints. In my AS/RS publication experience, I also worked with literature and validation related to multi-agent path planning, cooperative scheduling, and conflict resolution. This paper extends these ideas from simulated or scheduling-oriented settings to physical multi-robot systems.

Another important takeaway is that learning-based robotics often needs classical control support. The hybrid RL framework is especially meaningful to me because my Fast Robots project also relied on practical combinations of estimation, PID control, state-machine logic, and safety-oriented debugging. In real robot systems, it is often not enough to use either a learned policy or a traditional controller alone; robust deployment may require combining them.

5 Paper 3: Semantics-Aware Receding Horizon Planner for Object-Centric Active Mapping

Core Idea

The third paper studies object-centric active mapping. Traditional active mapping mainly focuses on geometric exploration: distinguishing occupied, free, and unknown spaces. However, many robotic tasks require semantic understanding. A robot may need to find target objects, inspect task-relevant structures, or build a map that contains object-level information rather than only obstacles.

The paper proposes a Semantics-Aware Receding Horizon Planner (SARHP). The system uses

RGB-D observations, object detection, and segmentation to generate semantic point clouds and build an object-centric volumetric map. The map is divided into three layers: a global mapping layer, a local mapping layer, and a Field-of-View mapping layer. This multi-layer design reduces the computational cost of frontier detection and route evaluation.

The planner evaluates candidate routes rather than only single viewpoints. It combines geometric information gain, semantic information gain, and kinematic cost. The geometric entropy of a voxel is written as

$$H_g(v) = -P_o(v) \ln P_o(v) - (1 - P_o(v)) \ln(1 - P_o(v)),$$

while the semantic entropy is written as

$$H_s(v) = -P_s(v) \ln P_s(v) - (1 - P_s(v)) \ln(1 - P_s(v)).$$

By combining these terms, the robot can choose routes that not only improve map coverage but also prioritize target-object understanding.

Takeaway

This paper helped me understand active perception as a planning problem. In Fast Robots, I implemented yaw-controlled 360-degree ToF scanning and grid-based localization. That experience showed me that the robot’s motion affects what it can observe and how well it can localize. SARHP takes this idea further: the robot actively selects routes and viewpoints to improve semantic map quality.

This paper also connects to my YOLOv11 project. My previous work focused on dense-scene object detection and improving visual perception performance. SARHP shows how object detection and semantic segmentation can be integrated into a robotic planning loop. Instead of treating perception as a standalone module, semantic information directly influences where the robot moves next.

6 Paper 4: Language-Augmented Symbolic Planner for Open-World Task Planning

Core Idea

The fourth paper studies open-world task planning. Long-horizon robot tasks require a system to decompose a high-level goal into executable actions. Traditional symbolic planners are interpretable and logically structured, but they require complete domain knowledge. In open-world environments, action preconditions, object properties, and available objects may be incomplete or unknown. Large language models provide commonsense reasoning, but they are unreliable as direct long-horizon

planners because they may generate invalid or ungrounded action sequences.

The paper proposes a Language-Augmented Symbolic Planner (LASP), which uses a large language model to repair incomplete symbolic knowledge rather than replacing the symbolic planner. The planning problem can be expressed as

$$P = (O, R, s_0, g, A),$$

where O is the object set, R is the predicate or property set, s_0 is the initial state, g is the goal, and A is the action set. Each action is represented by preconditions, add effects, and delete effects:

$$a = (\text{Pre}(a), \text{Add}(a), \text{Del}(a)).$$

When execution fails, LASP uses the LLM to diagnose the cause, generate missing preconditions, evaluate whether the existing predicate set can express them, translate natural language into PDDL, and add missing objects if necessary. The symbolic planner then replans using the updated knowledge base.

Takeaway

My main takeaway is that language models can be useful in robotics when they are used as structured reasoning assistants rather than unconstrained action generators. LASP keeps the symbolic planner as the main planning engine and uses the LLM to repair missing knowledge. This makes the system more interpretable than a purely language-driven planner.

This connects to my Overcooked-AI project because both involve long-horizon embodied decision-making. In Overcooked-AI, agents must decompose tasks into navigation, object handling, cooking, and delivery actions. LASP studies a more general and open-world version of this problem. It also connects to my Raspberry Pi-based embedded AI system, where I implemented voice-assisted command parsing, state management, and conflict resolution among different control modes. LASP helped me see how high-level language understanding can be connected with executable planning in embodied systems.

7 Paper 5: BiKC+: Bimanual Hierarchical Imitation With Keypose-Conditioned Coordination-Aware Consistency Policies

Core Idea

The fifth paper studies bimanual multi-stage manipulation. Many industrial and service tasks require two robot arms to coordinate across multiple stages, such as packing tools, hanging gar-

ments, inserting objects, or manipulating deformable materials. These tasks are difficult because they require both temporal synchronization and spatial coordination.

BiKC+ proposes a hierarchical imitation learning framework. A high-level keypose predictor forecasts the next target keypose, and a low-level trajectory generator produces action sequences conditioned on the predicted keypose. A keypose represents a sub-goal in a long-horizon manipulation task. The high-level predictor can be written as

$$\hat{k}_{j+1}, \hat{m}_{j+1} = g_{\omega}(o_t, k_j),$$

where $\hat{k}_{j+1}$ is the predicted next keypose and $\hat{m}_{j+1}$ indicates whether the next stage requires bimanual coordination. The low-level policy is conditioned on this keypose:

$$a_t = \pi_{\theta}(o_t, \hat{k}_{j+1}).$$

A major contribution of the paper is its coordination-aware keypose identification strategy. In coordination mode, the two arms must synchronize around shared goals. In non-coordination mode, each arm can preserve its own flexibility. The paper also uses consistency models for one-step action generation, which improves inference speed compared with diffusion-based policies.

Takeaway

Although my previous projects focused more on mobile robots and simulated embodied agents than on bimanual manipulation, this paper is still relevant to my research preparation. My Overcooked-AI project showed me that long-horizon embodied tasks often need sub-goals, role coordination, and temporal organization. BiKC+ studies a related idea in a more physically grounded manipulation setting.

This paper broadened my understanding from navigation-centered autonomy to manipulation-centered robot learning. It also showed that successful robot learning is not only about choosing a powerful model; it also requires the right task structure. In BiKC+, keyposes provide this structure by dividing a complex task into meaningful sub-stages. This is useful for thinking about general autonomous systems, where long-horizon behavior often needs hierarchical organization.

8 Integrated Understanding Across the Five Papers

After reading these five papers, I understand autonomous robotic systems as an integrated perception–planning–control problem rather than a single algorithmic task. A reliable robot should be able to answer five connected questions.

First, how can the robot convert raw observations into safe motion? NeuPAN addresses this question by using raw point observations for direct navigation while preserving model-based structure.

Second, how can multiple robots coordinate without centralized control? The distributed multi-robot collision-avoidance paper addresses this question by learning a decentralized sensor-level policy and combining it with hybrid control.

Third, how can the robot decide where to move in order to understand the environment better? SARHP addresses this question by integrating semantic information into active mapping and receding horizon planning.

Fourth, how can a robot plan long-horizon tasks when its symbolic knowledge is incomplete? LASP addresses this question by using language models to repair missing symbolic knowledge during execution.

Fifth, how can a robot learn coordinated long-horizon manipulation from demonstrations? BiKC+ addresses this question by using keypose-conditioned hierarchical imitation learning for bimanual tasks.

These five papers can be summarized as the following pipeline:

Raw Perception $\rightarrow$ Navigation and Collision Avoidance $\rightarrow$ Active Semantic Mapping $\rightarrow$ Task-Level Planning $\rightarrow$ C

This pipeline is highly relevant to my research preparation because it connects my practical experience in embedded robotics with broader research problems in autonomous systems. My prior projects gave me hands-on experience with noisy sensors, state estimation, real-time control, multi-agent coordination, and robot debugging. These readings help me reinterpret those experiences through the lens of learning-based robotic autonomy.

9 Connection to My Background and Future Direction

My experience in Fast Robots showed that reliable autonomy depends on a chain of modules: IMU and ToF sensing, Kalman filtering, PID control, state-machine logic, mapping, localization, BLE logging, and offline analysis. A failure in one part can affect the entire behavior. This experience connects most directly to NeuPAN and SARHP, which both study how sensing and planning can be integrated more tightly for robot navigation and mapping.

My Overcooked-AI project exposed me to multi-agent coordination, role assignment, task allocation, and shared-space constraints. This connects to the distributed collision-avoidance paper, LASP, and BiKC+. These papers show different levels of coordination: local motion coordination among multiple robots, high-level task planning under incomplete knowledge, and physical coordination between two robot arms.

My AS/RS publication experience gave me an initial understanding of multi-agent path planning and conflict resolution in warehouse-like systems. This background makes the distributed multi-robot navigation paper especially meaningful, because it studies a more physically grounded version of multi-agent conflict avoidance.

My YOLOv11 and embedded AI projects also support this reading direction. YOLOv11 connects with semantic perception in SARHP, while the Raspberry Pi-based embedded AI system connects with LASP through command interpretation, state management, and mode conflict resolution.

A possible future direction is to develop lightweight autonomous systems that combine active perception, robust local navigation, and task-level reasoning. For example, a small mobile robot could use sparse ranging and visual perception to build a semantic map, use a learning-based local controller to navigate safely, and use symbolic or language-augmented planning to decide task-level actions. Another direction is to study how robot logs can be used to identify weak operating conditions and improve future policy or planner design.

10 Potential Research Questions

Based on the five papers, I am interested in the following research questions:

1. How can direct point-based navigation methods be adapted to small embedded robots with sparse ToF, IMU, or low-cost depth sensing?
2. How can learning-based local navigation be combined with classical control to improve safety and deployability on real robots?
3. How can active semantic mapping be performed efficiently on resource-constrained mobile robot platforms?
4. How can language-augmented symbolic planning be connected with low-level robot navigation and manipulation policies?
5. How can multi-agent coordination ideas from simulated environments be transferred to physical multi-robot systems?
6. How can keypose or sub-goal structures be used beyond bimanual manipulation, such as in mobile manipulation or collaborative embodied tasks?

11 Conclusion

This reading project provided a structured understanding of learning-based autonomous robotic systems. The five papers together show that robot autonomy requires more than a single perception model, planner, or controller. It requires an integrated system that can perceive geometric and semantic information, navigate safely, coordinate with other agents, reason about long-horizon

tasks, and execute physical actions reliably.

For my future research preparation, the most important insight is that autonomous robotics should be treated as an integrated perception–planning–control problem. My background in embedded robot control, Kalman-filter-based state estimation, sensor fusion, multi-agent decision-making, and robot learning provides a practical foundation for this direction. Building on these readings, I am particularly interested in learning-based robot navigation, active perception, multi-agent coordination, and embodied decision-making for autonomous robotic systems operating in complex physical environments.

References

- [1] R. Han, S. Wang, S. Wang, Z. Zhang, J. Chen, S. Lin, C. Li, C. Xu, Y. C. Eldar, Q. Hao, and J. Pan, “NeuPAN: Direct Point Robot Navigation With End-to-End Model-Based Learning,” *IEEE Transactions on Robotics*, vol. 41, pp. 2804–2824, 2025.
- [2] T. Fan, P. Long, W. Liu, and J. Pan, “Distributed Multi-Robot Collision Avoidance via Deep Reinforcement Learning for Navigation in Complex Scenarios,” *The International Journal of Robotics Research*, vol. 39, no. 7, pp. 856–892, 2020.
- [3] L. Lu, Y. Zhang, P. Zhou, J. Qi, Y. Pan, C. Fu, and J. Pan, “Semantics-Aware Receding Horizon Planner for Object-Centric Active Mapping,” *IEEE Robotics and Automation Letters*, vol. 9, no. 4, pp. 3838–3845, 2024.
- [4] G. Chen, L. Yang, R. Jia, Z. Hu, Y. Chen, W. Zhang, W. Wang, and J. Pan, “Language-Augmented Symbolic Planner for Open-World Task Planning,” arXiv preprint arXiv:2407.09792, 2024.
- [5] H. Xu, Y. Chen, D. Yu, Y. Ren, and J. Pan, “BiKC+: Bimanual Hierarchical Imitation With Keypose-Conditioned Coordination-Aware Consistency Policies,” *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 1064–1079, 2026.